

Electrical and optical properties of materials

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Part 2: Dielectric properties of materials

In this, the second part of the course, we will examine the properties of dielectric materials, how they may be characterised, and how these characteristics depend on parameters such as temperature, and frequency of applied field. Finally, we shall review the ways in which dielectric materials fail under very high electric fields.

2.1 Ways to characterise dielectric materials

- a. Relative permittivity, ϵ_r
- b. Loss tangent, $\tan \delta$
- c. Breakdown field

2.1.1 Relative permittivity, ϵ_r

Let's begin by reminding ourselves about relative permittivity, which was introduced in earlier courses. Michael Faraday discovered that upon placing a slab of insulator between two parallel plates the charge on the plates increased, for a given voltage. This additional charge arises from an induced polarisation in the dielectric material.

Recap	$Q=CV$	$I = \frac{dQ}{dt} = C \frac{dV}{dt}$
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The capacitance increased with this insulator in place, such that we can define the new capacitance $C = \epsilon_r C_0$, where C_0 is the capacitance of the parallel plates when filled with a vacuum. The factor by which the capacitance increases is thus the relative permittivity, ϵ_r .

The electric displacement field D arises from the combination of the applied electric field E and the polarisation of the material P , in the relation:

$$(2.1)$$

Assuming the polarisation is proportional to the applied field E in the relation $P = \chi_e \epsilon_0 E$, we can rewrite this as:

$$(2.2)$$

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Thus, the induced polarisation serves to increase the apparent electric field by a factor ϵ_r , which we can now express as:

$$(2.3)$$

2.1.2 Loss tangent, $\tan \delta$

If we apply an AC voltage $V = V_0 \sin \omega t$ to a capacitor C , the current $I_C = V_0 \omega C \cos \omega t$ follows $\pi/2$ out of phase. The same will hold for a circuit containing purely capacitive components, as these can simply be expressed with an effective capacitance. The power lost in the circuit is the product of the current and voltage $W = \overline{IV}$, which is zero because I and V are precisely out of phase. Conversely, the current through a resistor R will follow the AC voltage in phase ($I_R = V_0 \sin \omega t / R$) and it dissipates power.

If the capacitor is ‘leaky’ in some way, such that there is a residual resistance, or the polarisation of the dielectric lags behind the AC voltage such that I and V are no longer perfectly out of phase, power will be lost across the capacitance. We define this characteristic in terms of the *loss tangent*. We model the leaky dielectric as a perfect capacitor with a resistor in parallel,

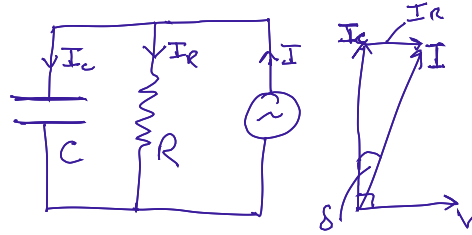


Figure 2.1: A leaky capacitor

as shown in Figure 2.1, and apply an AC voltage $V = V_0 \sin \omega t$.

$$(2.4)$$

We define the loss tangent $\tan \delta$ as the ratio of the amplitude of these components, such that a perfect capacitor has a loss tangent of zero.

$$(2.5)$$

The power lost $W = \overline{IV}$ is:

$$(2.6)$$

$$(2.7)$$

$$(2.8)$$

So there is power dissipation proportional to the loss tangent ($\tan \delta$) as well as relative permittivity ϵ_r . We sometimes use the *loss factor* ($\epsilon_r \tan \delta$) to compare dielectric materials by their power dissipation.

Another way of thinking about this is allowing a *complex* relative permittivity which incorporates this loss tangent. The imaginary part of ϵ_r is then directly responsible for the effective resistance. The impedance of a capacitor C is (C_0 is the capacitance of the device were it filled with vacuum):

$$(2.9)$$

For a complex $\epsilon_r = \text{Re}(\epsilon_r) + i\text{Im}(\epsilon_r)$:

$$(2.10)$$

The first of these terms is imaginary and so still looks like an ideal capacitor, with actual capacitance C' , while the second is real and so looks like a resistor ($Z = R$), as defined below:

$$(2.11)$$

The loss tangent is then nothing more than the ratio of the imaginary and real parts of the relative permittivity. Looking back at Eq. 2.3 we see the relationship between the relative permittivity and the polarisation induced in the dielectric. If the polarisation change is in phase with the applied electric field, the material appears purely capacitive. If there is a lag (for reasons we shall discuss in the coming section), the relative permittivity acquires some imaginary component, the material acquires ‘resistive’ character and a non-zero loss tangent.

2.2 Origins of polarisation

2.2.1 Electronic polarisation

Following the simple Bohr model of the atom, the applied electric field displaces the electron orbit slightly (see Figure 2.2). This produces a dipole,

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equivalent to a polarisation. There are quantum mechanical treatments of this effect (using perturbation or variational theory) which all give the result that the effect is both small, and occurs very rapidly (on a timescale equivalent to the reciprocal of the frequency of the X-ray or optical emission from excited electrons in those orbits). Therefore we expect no lag, and thus no loss, except at frequencies which are *resonant* with the electron transition energies. We will discuss the resonant case later.

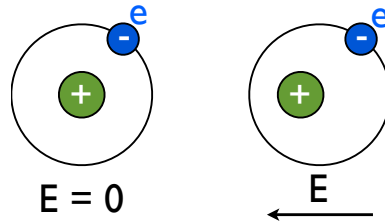


Figure 2.2: Electronic (atomic) polarisation

2.2.2 Ionic polarisation

The ions of a solid may be modelled as charged masses connected (to a first approximation) to their nearest neighbours by springs of various strengths, as illustrated in Figure 2.3. The electric field displaces the ions, polarising the solid. In this case we expect a profound frequency dependence on the lag, according to the charges, masses and ‘spring constants’ (interatomic forces).

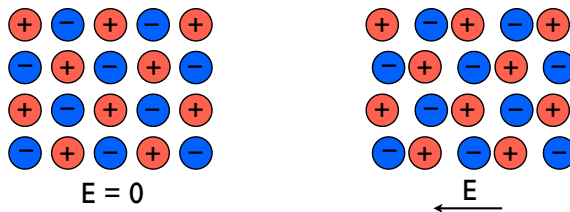


Figure 2.3: Ionic polarisation

2.2.3 Orientation polarisation

i. Fluids containing permanent electric dipoles: polar dielectrics

If the molecules of the fluid have permanent electrostatic dipoles, they will align with the applied electric field, as illustrated in Figure 2.4. Their be-

haviour is analogous to the classical theory of paramagnetism, which is examined in more detail in the following lecture course on Magnetic Properties of Materials. We shall simply note now that this leads to a $1/T$ temperature dependence. We may use intuition to observe that at low frequencies the molecules have time to respond to the applied electric field and so the polarisation can be large. On the other hand, if we apply a high frequency electric field, the molecules may not have time to respond by virtue of their inertia, collisions with other molecules etc., and so the polarisation can be less.

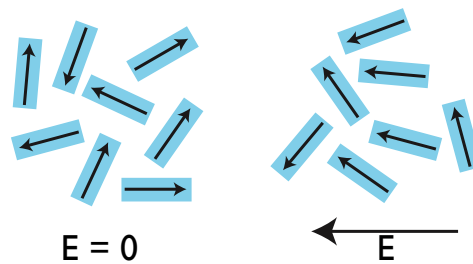


Figure 2.4: Polarisation due to electric dipoles in a fluid

ii. Ion jump polarisation

Dipoles across several ions in an ionic solid may reorient under an applied electric field to yield a net polarisation. For example, consider A^+B^- ionic solids containing a small amount of $C^{2+}(B^-)_2$ impurity. The A^+ vacancies which are present may associate with the C^{2+} (for net charge neutrality), and this pair will possess an electric dipole. This pair may reorientate under the applied field, through site-to-site changes of state, in order to minimise its energy, as illustrated in Figure 2.5. Both temperature and frequency dependencies are expected. The contribution will be small for frequencies much greater than the ion hopping frequency (as described in Part 1 of this course). This mechanism also applies in several of the models for ionic conductivity we examined earlier in which electric dipoles are present in the ionic solid.

2.2.4 Space charge polarisation

In a multiphase solid where one phase has a much larger electrical resistivity than the other, charges can accumulate at the phase interfaces. The material behaves like an assembly of resistors and capacitors on a fine scale, the overall effect being that the solid is polarised (a schematic drawing is shown

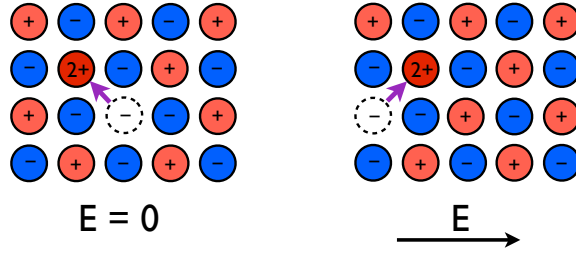


Figure 2.5: ‘Ion-jump’ polarisation

in Figure 2.6). A complicated frequency dependence is expected according to the range of effective capacitances and resistances involved, determined by grain sizes and the resistivity of the different phases (which are in turn temperature dependent). This type of polarisation can be observed in certain ferrites and semiconductors.

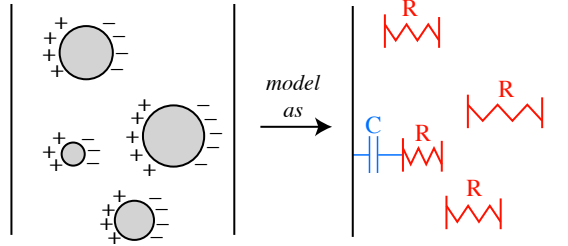


Figure 2.6: Space charge polarisation

In the following sections we will examine the frequency dependences of these different mechanisms. Figure 2.12 (towards the end of these notes) shows a basic summary, which should be consistent with our intuition on the energy/time scales of these processes.

2.3 Local electric field and polarisation

In order to understand the frequency response of these polarisation mechanisms, we must first develop a microscopic theory of polarisation and understand how the polarisability of particles in a certain effective field relates to measurable parameters such as ϵ_r . The polarisation induced at some point in our dielectric material is proportional to the *local* electric field:

$$(2.12)$$

where n is the density of particles of polarisability α . But how can we calculate the local electric field at some point, when this will itself include

contributions from the polarisation of the surrounding material? Let's take some point in the middle of the material. In the case of cubic symmetry, we can cut a spherical hole around that point and represent the effect of the polarisation of the material we've cut out by the surface charge resulting on the surface of the hole (see Figure 2.7). An electric field in the direction r

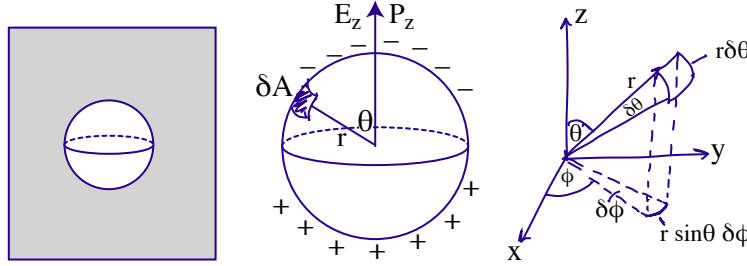


Figure 2.7: Calculation of the local electric field, by removing a sphere of material around the point of interest and finding the surface charge around the hole

is produced at the centre of the sphere by a small surface area of sphere δA with polarisation P (along z):

$$(2.13)$$

By symmetry, when we sum all the contributions of this radial electric field at the centre, only the component parallel to the polarisation (along z) will remain. Thus

$$(2.14)$$

Using the standard approach to surface integrals in spherical polar coordinates, where $\delta A = r^2 \sin \theta \delta \theta \delta \phi$, we have:

$$(2.15)$$

$$(2.16)$$

$$(2.17)$$

$$(2.18)$$

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(2.19)

The total electric field at some point within the material is the sum of this contribution from the surrounding polarisation, and the applied electric field E :

(2.20)

(Note: this derivation assumed cubic symmetry, and the constant prefactors in the above change when moving to other symmetries.) By combining this result with Eqs 2.3 and 2.12 we obtain¹ the *Clausius-Mossotti* relation:

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{n\alpha}{3\epsilon_0} \quad (2.21)$$

This relation reveals how a *microscopic* property of a material, the polarisability α , may be obtained from a measurable quantity ϵ_r .

2.3.1 Polarisability versus temperature

The term α in the Clausius Mossotti relation derived above represents how the overall polarisation of a material goes with the effective local (or internal) field. There must, however, be some temperature dependence — we can imagine that at high temperatures kinetic energies are such that the particles pay little attention to the electric field and the resulting polarisation is weak (and vice versa at low temperatures). We can use the *Langevin* derivation (developed originally for paramagnetism) to describe this effect.

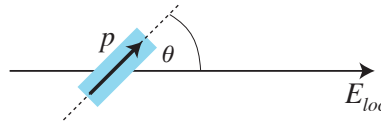


Figure 2.8: A permanent dipole in an electric field

Consider a particle with a permanent dipole p inclined at some angle θ to the *local* electric field E_{loc} (as illustrated in Figure 2.8. It has an electric potential energy $U = -pE_{loc} \cos \theta$. From classical statistical mechanics, the number of particles δn with energy in the range U to $U + \delta U$ is:

(2.22)

¹Tip: Start from $n\alpha = P/E_{loc}$ and evaluate right hand side

C is a normalising constant which ensures that $\int dn = n$. Our particle at angle θ to E_{loc} contributes $p \cos \theta$ to the overall polarisation (random components perpendicular to E_{loc} will cancel out on average). Hence, the volume polarisation is:

(2.23)

$$P = n \frac{\int_0^\pi p \cos \theta C \exp\left(\frac{pE_{loc} \cos \theta}{k_B T}\right) dU}{\int_0^\pi C \exp\left(\frac{pE_{loc} \cos \theta}{k_B T}\right) dU} \quad (2.24)$$

Differentiating U with respect to θ tells us: $dU = pE_{loc} \sin \theta d\theta$, and so:

$$P = n \frac{\int_0^\pi p \cos \theta C \exp\left(\frac{pE_{loc} \cos \theta}{k_B T}\right) pE_{loc} \sin \theta d\theta}{\int_0^\pi C \exp\left(\frac{pE_{loc} \cos \theta}{k_B T}\right) pE_{loc} \sin \theta d\theta} \quad (2.25)$$

This can be tackled by first cancelling the factor $pE_{loc}C$ from top and bottom and then making some handy substitutions $pE_{loc}/k_B T = y$ and $\cos \theta = x$ (which means $dx = -\sin \theta d\theta$):

$$P = n \frac{\int_1^{-1} x \exp(xy) dx}{\int_1^{-1} \exp(xy) dx} \quad (2.26)$$

This integral has a known solution:

(2.27)

$L(y)$ is known as the *Langevin function* and is plotted in Figure 2.9. In the limit of small y (small fields and/or high temperatures), $L(y) \rightarrow y/3$, i.e.

(2.28)

Looking back at Eq. 2.12, we now see that in this limit, the polarisability α has a $1/T$ dependence with temperature, and goes with the dipole *squared*. At the other limit (very high fields and/or low temperatures), all the dipoles add up and the polarisation becomes bounded to $P = np$. (Note that in this limit there is no further electric field dependence and Eq. 2.12 no longer holds).

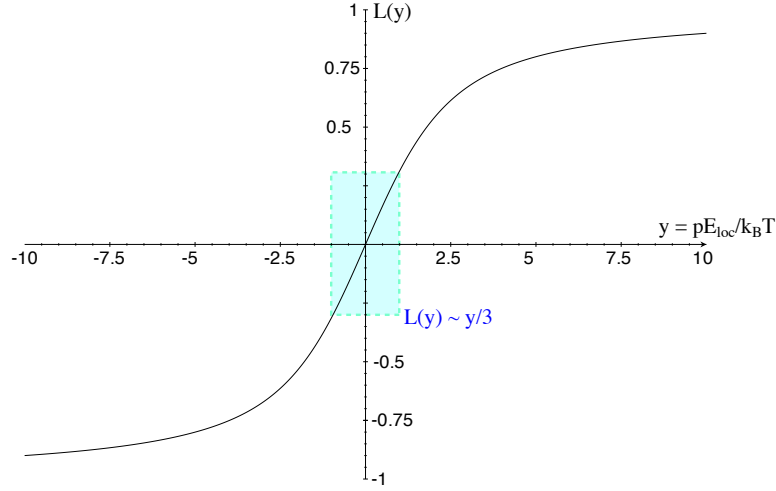


Figure 2.9: The Langevin function

2.4 Resonant frequency dependence ϵ_r

Many of the polarisation mechanisms described above can be thought of as some displacement of a charged particle from some equilibrium position. We can model this as a particle of charge q , mass m , held in the equilibrium by a force which is linear in displacement (i.e. a spring) with spring constant $m\omega_0^2$. Let's say the medium in which the particle sits provides a drag of constant $m\gamma$, and the particle experiences a force from an *oscillating* electric field $E_{loc} = E_0 \exp(i\omega t)$. The resulting equation of motion is:

$$(2.29)$$

We can guess that the steady state solution will be of the form $x_0 \exp(i\omega t)$, substitute such a solution into the differential equation above to check it works and find the constant x_0 .

$$x = x_0 \exp(i\omega t) = \frac{q}{m} \frac{1}{(\omega_0^2 - \omega^2) + i\omega\gamma} E_0 \exp(i\omega t) \quad (2.30)$$

The dipole moment of this particle is the product of its charge and its displacement: qx , so the total polarisation is:

$$P = np = nqx = \frac{nq^2}{m} \frac{1}{(\omega_0^2 - \omega^2) + i\omega\gamma} E_0 \exp(i\omega t) \quad (2.31)$$

Using Eq. 2.12 we can extract the polarisability:

$$\alpha = \frac{q^2}{m} \frac{1}{(\omega_0^2 - \omega^2) + i\omega\gamma} \quad (2.32)$$

and then use the Clausius-Mossotti relation (Eq. 2.21) to obtain the relative permittivity:

$$(2.33)$$

We notice that ϵ_r is therefore complex. The imaginary part is directly due to the ‘drag’ factor γ and leads to absorption of energy by the system, as we might expect.

2.4.1 Case for weak absorption

In the limit of weak resonance (ϵ_r close to 1), the Clausius-Mossotti relation simplifies to:

$$\frac{n\alpha}{3\epsilon_0} = \frac{\epsilon_r - 1}{\epsilon_r + 2} \approx \frac{\epsilon_r - 1}{3} \quad (2.34)$$

Thus we can write the frequency dependent ϵ_r derived in Eq. 2.33 as:

$$\epsilon_r - 1 = \frac{nq^2}{m\epsilon_0} \frac{1}{(\omega_0^2 - \omega^2) + i\omega\gamma} \quad (2.35)$$

and separate out the real and imaginary parts:

$$\text{Re}(\epsilon_r) = 1 + \frac{nq^2}{m\epsilon_0} \frac{(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2} \quad (2.36)$$

$$\text{Im}(\epsilon_r) = -\frac{nq^2}{m\epsilon_0} \frac{\omega\gamma}{(\omega_0^2 - \omega^2)^2 + \omega^2\gamma^2} \quad (2.37)$$

These expressions can be simplified for the case where the ω is close to the resonance frequency ω_0 with the substitution²: $(\omega_0^2 - \omega^2) = 2\omega(\omega_0 - \omega)$.

$$\text{Re}(\epsilon_r) = 1 + \frac{nq^2}{2m\omega\epsilon_0} \frac{(\omega_0 - \omega)}{(\omega_0 - \omega)^2 + \gamma^2/4} \quad (2.38)$$

$$\text{Im}(\epsilon_r) = -\frac{nq^2}{2m\epsilon_0} \frac{\gamma/2}{(\omega_0 - \omega)^2 + \gamma^2/4} \quad (2.39)$$

These terms are plotted in Figure 2.10 and show the maximum absorption (imaginary part of ϵ_r) right on resonance at ω_0 , as expected. For the limits where ω is small ($\omega \rightarrow 0$) or large ($\omega \rightarrow \infty$) we can go back to Eq. 2.35:

$$\text{If } \omega \rightarrow 0, \text{ then } \epsilon_r - 1 \rightarrow \frac{nq^2}{m\epsilon_0\omega_0^2} \quad (2.40)$$

²Let $\omega = \omega_0 + \delta$, and write down $(\omega_0^2 - \omega^2)$ neglecting powers of δ^2

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$$\text{If } \omega \rightarrow \infty, \text{ then } \epsilon_r - 1 \rightarrow 0 \quad (2.41)$$

In both cases the relative permittivity is purely real. We can think of the low frequency limit as that in which the polarisation can easily keep up with the oscillating electric field, and so there is no loss. At the other extreme, i.e. very high frequencies, the polarisation simply has no chance of following the oscillating electric field and ignores it. There is therefore a drift in the ‘background’ (non-resonant) part of ϵ_r to lower values as frequency is increased, in addition to the resonance peak. We can also see from the figure that the linewidth of the resonance feature is equal to γ .

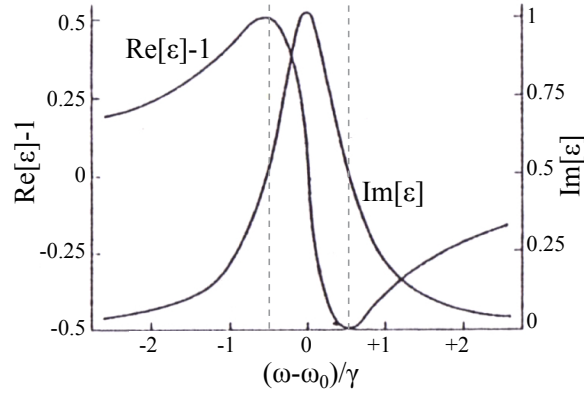


Figure 2.10: The real and imaginary parts of the relative permittivity close to resonance

We can naturally have multiple polarisation mechanisms/centres at play in a material, and might expect to see multiple resonances. This can be treated in the same way as above. Note that in the case of many resonances close together in frequency, discrete resonance lines may not be easily observed and care must be taken to correctly interpret the absorption spectrum.

The model we have used applies quite well to ionic solids, less well to molecular rotation bands, and surprisingly well to the optical/X-ray transitions in the electron polarisation model at very high frequencies.

2.5 *Non-resonant* frequency dependence of ϵ_r

In some mechanisms there will not be some particular resonance we are exciting, but rather there is a certain time response of the system to change (e.g. based on collisions in a fluid). This time response then determines the frequency behaviour of ϵ_r .

Let's consider the behaviour of a system with polarisation P under the influence of a field E , when the field suddenly changes to E_0 where the equilibrium polarisation would be P_0 . The change in polarisation won't be instantaneous, but will depend on the time response of the dominant polarisation mechanism. It is reasonable that the polarisation will exponentially tend to the equilibrium value with some rate constant τ .

(2.42)

We know that the frequency spectrum of something with a time dependence is given by the Fourier Transform (where constant A ensures $f(\omega)$ has the right dimension):

$$f(\omega) = A \int_0^\infty P(t) \exp(-i\omega t) dt \quad (2.43)$$

$$P(t) = P_0 \exp(-t/\tau) \quad (2.44)$$

(2.45)

(2.46)

(For complete generality we have added a constant offset B to this frequency dependence). Although this expression fully describes the frequency dependence of ϵ_r as it is, we commonly cancel the constants A and B by expressing $\epsilon_r(\omega)$ in terms of the static ($\omega = 0$) and high-frequency ($\omega = \infty$) limits. Evaluating Eq. 2.46 we see:

(2.47)

Thus we can write down the typical form of the *Debye Equation*

(2.48)

where, separating real and imaginary parts:

$$\text{Re}(\epsilon_r(\omega)) = \epsilon_r(\infty) + \frac{\epsilon_r(0) - \epsilon_r(\infty)}{1 + \omega^2\tau^2} \quad (2.49)$$

$$\text{Im}(\epsilon_r(\omega)) = \omega\tau \frac{\epsilon_r(0) - \epsilon_r(\infty)}{1 + \omega^2\tau^2} \quad (2.50)$$

These real and imaginary parts are sketched in Figure 2.11. We can see similarities with the resonant case described above. For example, we see a

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peak in the absorption (imaginary part) at some frequency $\omega = 1/\tau$. On the other hand, there is no resonant feature in the real part of ϵ_r as we saw in Figure 2.10, but rather a smooth change from $\epsilon_r(0)$ to $\epsilon_r(\infty)$. The model we've used describes well the behaviour of gases, and is thought to explain polar liquids and possibly ion-jump polarisation in solids.

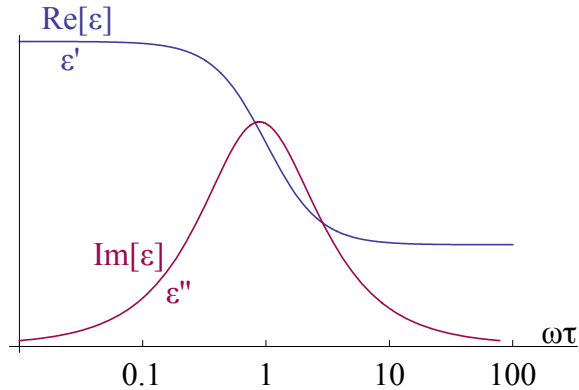


Figure 2.11: The real and imaginary parts of the relative permittivity close according to Debye relaxation

The different mechanisms and types of frequency dependence are summarised in Figure 2.12 and the table below. Finally, note that it is common to denote the real part of dielectric permittivity $\text{Re}(\epsilon_r) = \epsilon'_r$, and the imaginary part $\text{Im}(\epsilon_r) = \epsilon''_r$.

2.6 Temperature and frequency dependence in the loss factor

In general, there are two contributions to the loss factor:

- a. DC leakage which appears as a resistance R . Polar organic molecules are very low loss, while ionic solids (as we've discovered) have a strongly temperature dependent conductivity.
- b. AC loss arising from the imaginary part of the dielectric permittivity ϵ_r , which has the frequency dependences described above. For example, oils used in transformers to prevent breakdown are OK at very low frequencies, but at medium-to-high frequencies ionic solids must be used. Oils may have Debye-type losses in the medium frequency range, leading to a drop in ϵ_r which persists to infinite frequency.

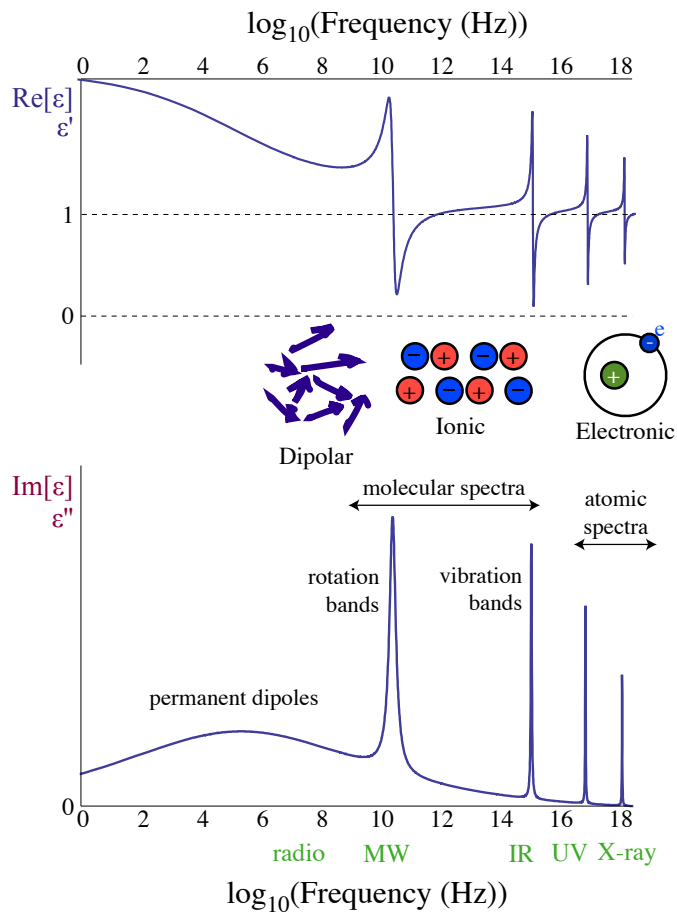


Figure 2.12: Summary of various polarisation mechanisms and how they contribute to the frequency dependence of ϵ_r (not all present in one material).

Polarisation	Temperature dep.	Frequency dep.
Electronic	none	very little, except at very high frequencies
Ionic	some, as spring constants vary with T	resonances in IR
Fluid orientation	$\propto 1/T$	Large - Debye
Ion jump	some: tends to 0 as $T \rightarrow 0$ (no jumps) and as $T \rightarrow \infty$ (K.E. $\gg$ orientation energy)	Debye
Space charge	Resistivities of different phases strongly T dependent	yes ('RC network')

2.7 Dielectric breakdown in semiconductors

This third property of dielectric materials may be treated somewhat independently from the other two. *Breakdown* describes the situation where, under very high electric fields, the material becomes conducting (lightning being a classic example). The electric field at which the dielectric eventually breaks down is known as the *dielectric strength* or *breakdown strength*.

As we learnt in Part 1, insulators such as ionic solids behave like semiconductors of large energy gap. There are two *non-catastrophic* breakdown mechanisms associated with semiconductors which might therefore be applicable: *Zener breakdown* and *avalanche breakdown*.

2.7.1 Zener breakdown

Take a look at Figure 2.13, where a large electric field is applied to a semiconductor. Typically, an electron in the valence band lacks the energy to enter the conduction band. However, in a strong electric field the energy bands bend with distance and an electron can hop from the valence to conduction band by changing its position. There is clearly an energy barrier to such a jump, but as the field increases, the necessary distance Δx decreases and there is increasing likelihood that the electron *tunnels* successfully. Note that it also leaves a hole behind, which will also conduct.

2.7.2 Avalanche breakdown

In even larger electric fields (a larger bandgap will require larger fields for Zener breakdown) an electron, once in the conduction band, may acquire very large amounts of kinetic energy in between collisions. It may be that when it does experience a collision, sufficient energy can be given to a valence-band electron to promote it into the conduction band. The original electron loses some kinetic energy but stays in the conduction band. There are now two electrons (and two holes) which can then generate more, producing an ‘avalanche’ effect.

2.8 Dielectric breakdown in insulators

Breakdown in insulators (where it is understood at all) is usually classified under the following five headings:

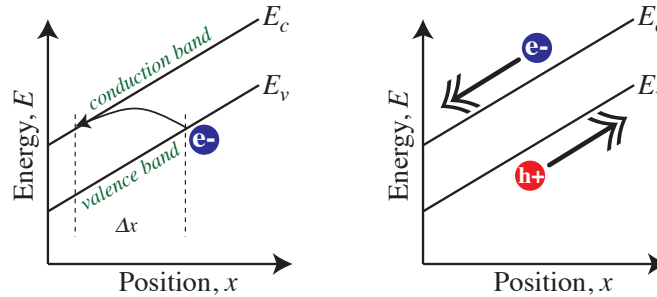


Figure 2.13: Zener breakdown (left) and collision, or avalanche, breakdown (right)

2.8.1 Collision breakdown

This is the same mechanism as avalanche breakdown: although in insulators the concentration of electrons in the conduction band is extremely weak ($\sim 10^6 \text{ m}^{-3}$). At high fields these few electrons acquire large amounts of kinetic energy and thus through collisions multiply the number of free electrons. [Note: Zener breakdown is much less likely in insulators because the tunnelling length becomes too large — a consequence of the large band gap]

2.8.2 Thermal breakdown

Under DC conditions, once any conduction starts to take place, ohmic heating will result. Because thermal conductivities of insulators are often very low, a large degree of local heating is possible. In insulators, as in semiconductors, higher temperature results in more free carriers and a higher conductivity. This can cause more conduction, more heating (even melting) as breakdown ensues.

Under AC conditions, any loss (lag) mechanism dissipates energy and hence heats the material, leading to breakdown as above. As an example, the Debye relaxation term for polythene has a maximum at 1 MHz. The molecule will absorb more strongly at this frequency, causing heating and a dramatically lower breakdown strength than at DC:

$$\begin{array}{ll} \text{Breakdown strength@ DC} & 3 - 5 \times 10^8 \text{ Vm}^{-1} \\ \text{Breakdown strength @ 1 MHz} & 5 \times 10^6 \text{ Vm}^{-1} \end{array}$$

2.8.3 Gas-discharge breakdown

Common lightning is an obvious example of this effect, though it may also be the dominant mechanism in a solid if the insulator is porous, containing oc-

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cluded gas bubbles (e.g. interlayer air in some micas). The field experienced by the gas is higher than that in the solid because of the continuity condition on the electric displacement field $D = \epsilon_r \epsilon_0 E$ (see Figure 2.14). Because $\epsilon_r(\text{solid})$ is likely to be significantly greater than $\epsilon_r(\text{gas})$, a larger electric field E will be present within the gas region. Thus, even if the breakdown strength of the gas were to be greater than that of the solid, it is likely to fail at a lower applied field because of the amplifying effect of the relative permittivity of the solid.

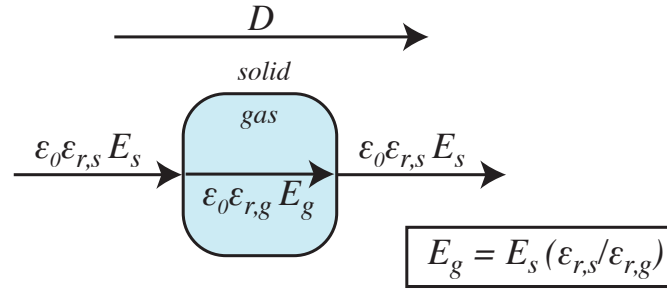


Figure 2.14: Gas discharge breakdown: the field-amplifying effect of the relative permittivity of the solid

2.8.4 Electrolytic breakdown

This term is used for breakdown caused by the presence of structural imperfections such as dislocation arrays, grain boundaries etc. which produce electrically weaker or conducting paths in the material through which current may pass.

2.8.5 Dipole breakdown

Related but distinct from the above mechanism is dipole breakdown, where structural imperfections stress the dipoles produced when the material becomes highly polarised in such a way as to make them more easily ionised. This increases the concentration of free carriers providing semiconducting paths with significantly lower electrical resistivity.

In an actual material, all of these mechanisms may be active to some extent. Thermal and collision breakdown are certainly ‘the last straw’, giving catastrophic breakdown in many cases.